An Efficient Harmonic Balance Method for Nonlinear Eddy Current Problems

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Abstract— A method is presented which combines the Fix-Point and the Harmonic-Balance Method. The potentials are represented by Fourier-series and the nonlinearity of the material is taken into account by the Fix-Point-Method. The equations for the unknown Fourier coefficients of the potentials are decoupled for each harmonic component. This method avoids calculating transient processes which normally have to be stepped through if using time-stepping methods.

I. Introduction

In nonlinear electromagnetic problems with periodic excitation the steady-state solution is often sufficient. With time-stepping methods a lot of periods may have to be stepped through until steady-state is achieved. For static fields an improvement can be made by using the *Fix-Point Method* [1]. This method can also be used for time periodic problems [2]. In this case, the equations for all time instants within a period are decoupled from each other. Another way for which the steady-state can be obtained easily is by introducing Fourier-series called the *Harmonic-Balance Method*. In [3], Fourier-series are used for the vector potential **A** and the magnetic reluctivity ν.

In this paper, the *Harmonic-Balance Method* and the *Fix-Point Method* are coupled. By using the Fix-Point Method, a Fourier-series for the reluctivity is not applicable but several iterations have to be made to take the nonlinearity into account. The unknowns are the Fourier coefficients of the introduced potentials **A** and v.

II. Method

In eddy current regions, the magnetic vector potential **A** and the time integrated electric scalar potential v can be introduced. The magnetic field intensity **H** is written as:

$$\mathbf{H} = \mathbf{H}(B) = \nu_{FP}\mathbf{B} - \mathbf{M}_{FP} \tag{1}$$

where ν_{FP} is a fixed value which influences the convergence of the method and $\mathbf{M}_{FP}$ is a magnetization-like quantity which includes the nonlinear behavior of the material. This approach leads to the following form of Ampere's law and the law of charge conservation:

$$\begin{aligned} &\nu_{FP}\operatorname{curl}\left(\operatorname{curl}\mathbf{A}\right) + \sigma\frac{\partial \mathbf{A}}{\partial t} + \sigma\operatorname{grad}\frac{\partial \mathrm{v}}{\partial t} = \operatorname{curl}\mathbf{M}_{FP} \\ &\operatorname{div}\left(\sigma\frac{\partial \mathbf{A}}{\partial t} + \sigma\operatorname{grad}\frac{\partial \mathrm{v}}{\partial t}\right) = 0 \end{aligned} \tag{2}$$

If Galerkin's Method is used an algebraic equation system can be obtained:

$$\mathbf{C}\cdot\tilde{\mathbf{A}}(t) + \mathbf{D}\cdot\dot{\tilde{\mathbf{A}}}(t) + \mathbf{E}\cdot\dot{\tilde{\mathbf{v}}}(t) = \mathbf{f}(t) \tag{3}$$

with constant matrices **C**, **D** and **E**. The time dependent unknown vectors $\tilde{\mathbf{A}}(t)$ and $\tilde{\mathbf{v}}(t)$ and the known vector **f**(t) are approximated as Fourier-series:

$$\begin{aligned} \tilde{\mathbf{A}}(t) &= \tilde{\mathbf{A}}_0 + \sum_{m=1}^{M}\tilde{\mathbf{A}}_{mc}\cos\left(m\omega_0 t\right) + \sum_{m=1}^{M}\tilde{\mathbf{A}}_{ms}\sin\left(m\omega_0 t\right) \\ \tilde{\mathbf{v}}(t) &= \tilde{\mathbf{v}}_0 + \sum_{m=1}^{M}\tilde{\mathbf{v}}_{mc}\cos\left(m\omega_0 t\right) + \sum_{m=1}^{M}\tilde{\mathbf{v}}_{ms}\sin\left(m\omega_0 t\right) \\ \mathbf{f}(t) &= \mathbf{f}_0 + \sum_{m=1}^{M}\mathbf{f}_{mc}\cos\left(m\omega_0 t\right) + \sum_{m=1}^{M}\mathbf{f}_{ms}\sin\left(m\omega_0 t\right) \end{aligned} \tag{4}$$

With these approximations, (3) can be separated in terms of cos(mω$_0$t) and sin(mω$_0$t) and a time independent term. So, M+ 1 separate equation systems are obtained:

$$\begin{gathered} \mathbf{C}\cdot\mathbf{A}_0 = \mathbf{f}_0 \\ \begin{bmatrix} \mathbf{C} & m\omega_0\mathbf{D} & \mathbf{0} & m\omega_0\mathbf{E} \\ -m\omega_0\mathbf{D} & \mathbf{C} & -m\omega_0\mathbf{E} & \mathbf{0} \end{bmatrix} \cdot \begin{bmatrix} \tilde{\mathbf{A}}_{mc} \\ \tilde{\mathbf{A}}_{ms} \\ \tilde{\mathbf{v}}_{mc} \\ \tilde{\mathbf{v}}_{ms} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{mc} \\ \mathbf{f}_{ms} \end{bmatrix} \\ m = 1,\ldots,M \end{gathered} \tag{5}$$

III. The Algorithm

Starting from an arbitrary value of $\mathbf{M}_{FP}$, the vector **f**(t) is determined. With the solution of the equation system (5), a new value for $\mathbf{M}_{FP}$ can be determined from equation (1). This procedure will be repeated until the changes of **f**(t) between two iterations are smaller then a prescribed value.

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V. References

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